

Student: Ondřej Marvan (477 001)
Class time: Thursday 15:00

Project Title: **Chess Game Simulation Engine**

Project Description:

The goal of this project is to build a reusable and extensible chess game simulation engine in R, packaged as a standalone R package with an interactive Shiny dashboard.

The engine will allow users to simulate chess games between configurable AI players using different built-in strategies (e.g. random moves, greedy piece capture, basic minimax evaluation). Users will be able to run tournaments between multiple AI strategies, track results on a live leaderboard, and replay selected games move by move — all through an interactive Shiny dashboard.

The project is designed as a general-purpose simulation tool rather than a one-off analysis, with a clean architecture that makes it easy to extend with new player strategies or game variants.

Techniques to be applied (4 required for individual project):

1. Object-oriented programming (R6) — core game entities (Board, Piece, Player, Game) modelled as R6 classes with encapsulated state and methods
2. C++ via Rcpp — computationally intensive components such as move generation, board evaluation and game-tree search implemented in C++ for performance
3. Shiny application — interactive dashboard for configuring players and strategies, running simulations, viewing tournament leaderboards and replaying games
4. R package — the full engine structured and documented as an installable R package

Note regarding presentation: I am registered for the Thursday 15:00 group. **I would kindly ask not to be scheduled for the presentation on 14.05.2026, as I will be unable to attend that class.**